

Table 1. Summary MGSM multilingual-transfer results for Qwen3-4B and Llama-3.2-3B-Instruct, macro-averaged equally across five languages within each model. Higher is better for every column.

Model	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Original	41.6	0.00	1.76	1.47
	S-PID	0.0	<u>1.98</u>	0.00	0.00
	A-LQR	22.4	1.81	1.51	1.08
	H_∞ (ours)	32.2	1.44	1.66	1.25
Llama-3.2-3B	Original	33.0	0.00	1.68	1.44
	S-PID	8.2	<u>1.94</u>	0.87	0.60
	A-LQR	<u>27.6</u>	1.50	<u>1.57</u>	<u>1.17</u>
	H_∞ (ours)	25.8	0.83	1.46	1.08